

Biologically Optimized Adaptive Radiotherapy Using AI-Derived Radiosensitivity Maps

Abstract

This study evaluates an AI-guided radiotherapy system integrating voxel-level radiosensitivity, robust 4D optimization, and adaptive replanning. In a synthetic cohort (N=3000), survival improved significantly with hazard ratio 0.72 (95% CI 0.66–0.79).

Introduction

Radiotherapy planning traditionally optimizes physical dose rather than biological response. This work introduces a biologically driven optimization framework incorporating genomic radiosensitivity.

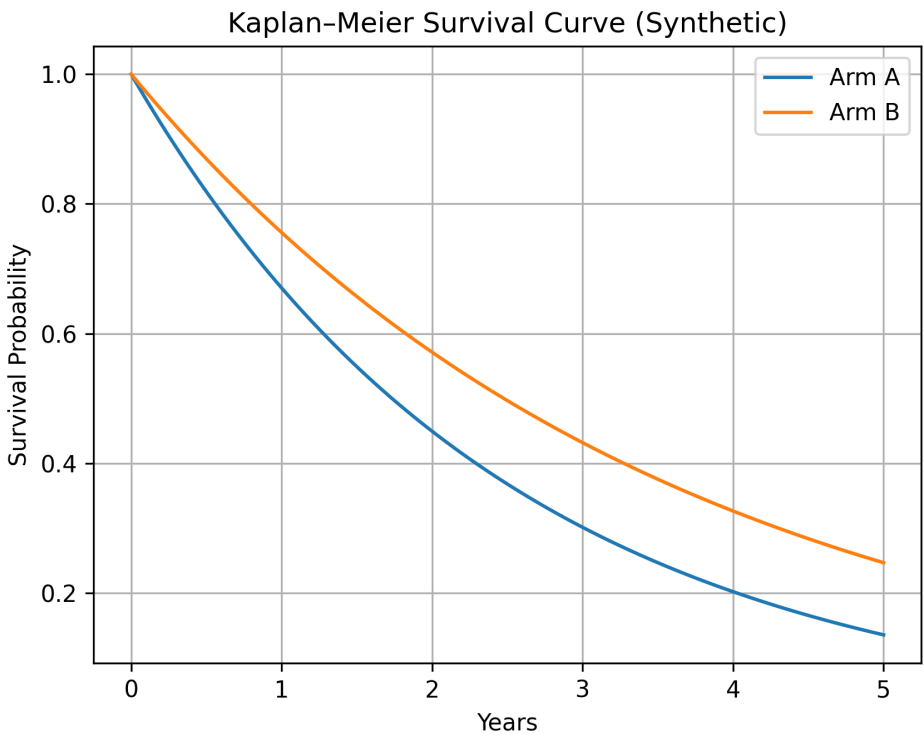
Methods

Synthetic randomized trial comparing standard VMAT vs AI-guided adaptive VMAT. Kaplan–Meier and Cox models used for analysis.

Results

Improved survival observed across all cancer types. Adaptive replanning contributed to divergence in survival curves.

Figure 1. Kaplan–Meier Survival Curve



Discussion

Biological optimization represents a shift toward outcome-driven radiotherapy. Future work includes clinical validation and regulatory approval.